

Calcular a área delimitada pelas funções $2x+1$, $x/2+4$ e $-x-2$:



$$f(x) = 2x + 1$$

$$g(x) = \frac{x}{2} + 4$$

$$h(x) = -x - 2$$

$$\frac{x}{2} + 4 = -x - 2$$

$$2x + 1 = -x - 2$$

$$\frac{x}{2} + 4 = 2x + 1$$

$$x + 8 = -x - 4$$

$$2x + x + 1 = -2$$

$$x + 8 = 4x + 2$$

$$x + 2x + 8 = -4$$

$$3x + 1 = -2$$

$$x - 4x + 8 = 2$$

$$3x + 8 = -4$$

$$\frac{3x}{3} = \frac{-12}{3}$$

$$-3x + 8 = 2$$

$$\frac{3x}{3} = \frac{-12}{3}$$

$$x = -4$$

$$-3x = -6$$

$$x = -4$$

$$\frac{3x}{3} = \frac{-6}{3}$$

$$x = -4$$

$$x = \frac{6}{3}$$

$$x = 2$$

$$-4 \vee -1 \vee 2$$

$$g(x) = \frac{x}{2} + 4 \quad [-4, -1]$$

$$h(x) = -x - 2$$

$$g(-2) = \frac{-2}{2} + 4 = -1 + 4 = 3$$

$$h(-2) = -(-2) - 2 = 2 - 2 = 0$$

$$\int_{-4}^{-1} \left[\left(\frac{x}{2} + 4 \right) - (-x - 2) \right] dx$$

$$f(x) = 2x + 1 \quad [1, 2]$$

$$g(x) = \frac{x}{2} + 4$$

$$f(0) = 2 \cdot 0 + 1 = 1$$

$$g(0) = \frac{0}{2} + 4 = 4$$

$$\int_{-1}^2 \left[\left(\frac{x}{2} + 4 \right) - (2x + 1) \right] dx$$

Integral 1

$$\int_{-4}^{-1} \left[\left(\frac{x}{2} + 4 \right) - (-x - 2) \right] dx$$

$$\left(\frac{x}{2} + 4 \right) = \frac{x}{2} + 4$$

$$-(-x - 2) = x + 2$$

$$\frac{x}{2} + 4 + x + 2$$

$$\frac{x}{2} + x + 6 =$$

$$\frac{x}{2} + \frac{x}{1} + 6 =$$

$$\frac{x}{2} + \frac{2x}{2} + 6 =$$

$$\frac{3x}{2} + 6$$

$$\int_{-4}^{-1} \left[\frac{3x}{2} + 6 \right] dx$$

$$\frac{3x}{2} \rightarrow \frac{3x^2}{4}$$

$$6 \rightarrow 6x$$

$$\left[\frac{3x^2}{4} + 6x \right]_{-4}^{-1}$$

Integral 2

$$\int_{-1}^2 \left[\left(\frac{x}{2} + 4 \right) - (2x + 1) \right] dx$$

$$\left(\frac{x}{2} + 4 \right) = \frac{x}{2} + 4$$

$$-(2x + 1) = -2x - 1$$

$$\frac{x}{2} + 4 - 2x - 1$$

$$\frac{x}{2} + 3 - 2x$$

$$\int_{-1}^2 \left(\frac{-3x}{2} + 3 \right) dx$$

$$\frac{x}{2} + 3 - \frac{4x}{2}$$

$$\frac{-3x}{2} + 3$$

$$\frac{-3x}{2} \rightarrow \frac{-3x^2}{4}$$

$$3 \rightarrow 3x$$

$$\left[\frac{-3x^2}{4} + 3x \right]_{-1}^2$$

Calcular Integrais

$$\left[\frac{3x^2}{4} + 6x \right]_{-4}^{-1} \quad \int_{-4}^{-1} \left[\left(\frac{x}{2} + 4 \right) - (-x - 2) \right] dx$$

$$\frac{3 \cdot (-1)^2}{4} + 6 \cdot (-1) = \frac{3}{4} + (-6) = \frac{3}{4} - \frac{24}{4} = -\frac{21}{4}$$

$$\frac{3 \cdot (-4)^2}{4} + 6 \cdot (-4) = \frac{3 \cdot 16}{4} + (-24) = \frac{48}{4} + (-24) = 12 - 24 = -12$$

$$-\frac{21}{4} - (-12) = -\frac{21}{4} + \frac{12}{1} = -\frac{21}{4} + \frac{48}{4} = \frac{27}{4}$$

$$\left[\frac{-3x^2}{4} + 3x \right]_{-1}^2 \quad \int_{-1}^2 \left(\frac{-3x}{2} + 3 \right) dx$$

$$\frac{-3 \cdot 2^2}{4} + 3 \cdot 2 = \frac{-3 \cdot 4}{4} + 6 = \frac{-12}{4} + 6 = \frac{-12}{4} + \frac{24}{4} = \frac{12}{4} = 3$$

$$\frac{-3 \cdot (-1)^2}{4} + 3 \cdot (-1) = \frac{-3 \cdot 1}{4} + (-3) = \frac{-3}{4} - 3 = \frac{-3}{4} - \frac{12}{4} = \frac{-15}{4}$$

$$3 - \left(\frac{-15}{4} \right) = 3 + \frac{15}{4} = \frac{3}{1} + \frac{15}{4} = \frac{12}{4} + \frac{15}{4} = \frac{27}{4}$$

$$\text{Área total: } \frac{27}{4} + \frac{27}{4} = \frac{54}{4} = \frac{27}{2}$$